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Bereken in $\mathbb{C}$ de vierkantswortel van de volgende complexe getallen: $1+2\sqrt{6}i$.
 $1 + 2\sqrt{6}i = (x + yi)^2$

$$\Leftrightarrow \begin{cases} x^2 - y^2 = 1 \\ 2xy = 2\sqrt{6} \end{cases}$$

$$\Leftrightarrow \begin{cases} x^2 - y^2 = 1 \\ xy = \sqrt{6} \end{cases}$$

$$\Leftrightarrow \begin{cases} x^2 - y^2 = 1 \\ x^2(-y)^2 = 6 \end{cases}$$

x^2 en y^2 zijn oplossingen van:

$$\lambda^2 - \lambda - 6 = 0$$

$$\Delta = 1 + 4 * 6 = 25$$

$$\lambda_{1,2} = \frac{1 \pm \sqrt{25}}{2}$$

$$\lambda_1 = 3 \text{ en } \lambda_2 = -2$$

$$x = \sqrt{3} \text{ en } y = \sqrt{2}$$

$$\text{Oplossing: } \sqrt{3} + \sqrt{2}i \text{ of } -\sqrt{3} - \sqrt{2}i$$

References